

Animal Kingdom

PU-II Biology • Chapter 4

Class Presentation Slides

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Basis of Classification

- Levels of organisation
- Symmetry
- Germ layers
- Coelom

Focus: Over a million species of animals have been described so far, making classification essential. Basis of classification includes: levels of organisation, patterns of digestive/circulatory/reproductive systems, b

Body Symmetry, Germ Layers and Coelom

- Radial
- Bilateral
- Asymmetry
- Diploblastic

Focus: radial symmetry: any plane through the central axis divides the body into identical halves (e.g. coelenterates, ctenophores, echinoderms). Bilateral symmetry: only one plane divides the body into identical left

Segmentation and Notochord

- Metamerism
- Chordates
- Non-chordates

Focus: Segmentation: body externally and internally divided into segments with repeated organs, called metameric segmentation or metamerism (e.g. Earthworm). Notochord is a mesoderm-derived rod-like structure formed d

Phylum Porifera

- Sponges
- Cellular level
- Canal system
- Ostia/Spongocoel/Osculum

Focus: Commonly called sponges; generally marine, primitive, multicellular, asymmetrical, with cellular level of organisation. Have a water transport / canal system: water enters through ostia into the spongocoel and

Phylum Coelenterata (Cnidaria)

- Cnidoblasts
- Polyp and medusa
- Metagenesis

Focus: Coelenterates/Cnidaria are aquatic (mostly marine), sessile or free-swimming, radially symmetrical, diploblastic, with tissue level of organisation. Have a central gastro-vascular cavity (GVC) with a single ope

Phylum Ctenophora

- Comb jellies
- Bioluminescence
- Comb plates

Focus: Commonly known as sea walnuts, comb jellies or sea gooseberries; exclusively marine, radially symmetrical, diploblastic, tissue level of organisation. Digestion is extracellular or intracellular; locomotion is

Phylum Platyhelminthes

- Flatworms
- Flame cells
- Hermaphrodite

Focus: Commonly called flatworms due to dorso-ventrally flattened bodies; bilaterally symmetrical, triploblastic, acoelomate, organ level of organisation. Nutrition is parasitic (endoparasites) with hooks and suckers

Phylum Aschelminthes

- Round worms
- Pseudocoelomates
- Dioecious

Focus: Known as round worms due to circular cross-section body; bilaterally symmetrical, triploblastic, pseudocoelomate, organ system level of organisation. May be free-living (aquatic/terrestrial) or parasitic on pla

Phylum Annelida

- Segmented worms
- Nephridia
- Closed circulation

Focus: May be aquatic (marine/freshwater) or terrestrial; bilaterally symmetrical, triploblastic, coelomate, organ system level, with metamerically segmented bodies. Locomotion by longitudinal and circular muscles; pa

Phylum Arthropoda

- Largest phylum
- Jointed appendages
- Open circulation
- Malpighian tubules

Focus: largest phylum in Animalia, comprising over two-thirds of known animal species, mostly insects; bilaterally symmetrical, triploblastic, coelomate (haemocoel — coelom filled with blood). Body covered by chitinous

Phylum Mollusca

- Second largest phylum
- Mantle cavity
- Radula

Focus: second largest phylum in Animalia among invertebrates; terrestrial or aquatic; bilaterally symmetrical, triploblastic, coelomate, organ system level. Body covered by a calcareous shell with distinct head, muscu

Phylum Echinodermata

- Spiny-bodied
- Water vascular system
- Marine

Focus: Echinodermata means spiny-bodied due to an endoskeleton of calcareous ossicles; exclusively marine; bilaterally symmetrical, triploblastic, coelomate, organ system level. Digestive system is complete with mouth

Phylum Hemichordata

- Worm-like marine animals
- Proboscis, collar, trunk
- Earlier placed as a sub-phylum of Chordata; a small group of worm-like marine animals; bilaterally symmetrical, triploblastic, coelomate, organ system level. Body is cylindrical, composed of anterior proboscis, collar and a long trunk.
- Circulatory system open; respiratory organs are gills; excretory organ is the proboscis gland. Sexes separate, fertilisation external, development indirect. Examples: Balanoglossus (tongue worm/acorn worm) and Saccoglossus.

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Phylum Chordata — General Features and Subphyla

- Notochord
- Dorsal nerve cord
- Gill slits
- Post-anal tail

Focus: Fundamental characters of Chordata: presence of notochord, dorsal hollow nerve cord, and paired pharyngeal gill slits. Bilaterally symmetrical, triploblastic, coelomate, organ system level; possess a post-anal

Class Cyclostomata

- Jawless
- Ectoparasitic
- Cartilaginous

Focus: Sucking circular mouth without jaws; elongated, eel-like body. Marine but migrate to fresh water for spawning; die shortly after spawning; larvae metamorphose and return to the ocean.

Class Chondrichthyes

- Cartilaginous fish
- Placoid scales
- Poikilothermous

Focus: Marine habitat; cartilaginous endoskeleton, streamlined body; notochord persists throughout life. Skin has small placoid scales; predaceous with strong jaws, mouth located ventrally, teeth are modified placoid

Class Osteichthyes

- Bony fish
- Operculum
- Air bladder

Focus: Marine and freshwater habitats; bony endoskeleton, streamlined body with terminal mouth. Body covered by cycloid or ctenoid scales; respiration by four pairs of gills covered by operculum; air bladder provides

Class Amphibia

- Water and land
- Cloaca
- Three-chambered heart

Focus: Can live both in water and on land; body divisible into head and trunk with two pairs of limbs; tail may be present in some. Alimentary canal, excretory and reproductive tracts open into a common cloaca; skin i

Class Reptilia

- Creeping/crawling
- Dry cornified skin
- Oviparous

Focus: Characterised by creeping/crawling locomotion; mostly terrestrial; body covered by dry cornified skin with epidermal scales/scutes. No external ears but tympanum present; two pairs of limbs when present; heart

Class Aves

- Wings and flight
- Four-chambered heart
- Homoiothermous

Focus: Characterised by wings and flight ability (except flightless birds like ostrich); have beaks; forelimbs modified into wings, hind limbs used for walking/swimming/clasping branches. Skin is dry with oil glands o

Class Mammalia

- Mammary glands
- Hair
- Viviparous

Focus: Found in a wide variety of habitats — polar ice, deserts, mountains, forests, grasslands, dark caves; adapted to fly, swim or walk. Presence of mammary glands (producing milk for young) is a unique mammalian ch

Summary — Salient Features of Phyla

- Quick comparison table
- This summary table brings together level of organisation, symmetry, coelom, segmentation, digestive/circulatory/respiratory systems, and distinctive features for all eleven phyla covered in this chapter — ideal for quick pre-exam revision.
- Use the salient comparisons in this topic to distinguish groups accurately in classification questions.

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